Why we can win DSN 2026

Team briefing · Find the Language I Am Speaking · Tuesday 20 October 2026, 12:30 Nigeria time (WAT)

In one line: the competition looks like a contest of big models, but it is really decided by four small languages (Efik, Ibibio, Igala, Nupe) that every team is getting almost completely wrong. The reason is a shortage of **different voices**, not model size, and that is what our plan fixes.

1. The competition in one minute

- **Task:** hear a 2–30 second clip and say which of 12 Nigerian languages it is (Yoruba, Igbo, Hausa, Eggon, Annang, Efik, Ibibio, Idoma, Igala, Nupe, Tiv, Urhobo), or **other** (English, Pidgin, anything else).
- **Score:** accuracy (a weighted F1) × an efficiency bonus between 0.5 and 1.0 for being small, fast and light. Yoruba, Igbo and Hausa count 4.3% each; **every other class counts double, 8.7%**.
- **How we submit:** we publish the model on Hugging Face and upload a small zip with its link. The organisers run it on a hidden test set of **new speakers** recorded in clean, noisy and **phone-quality** conditions.
- **Deadlines:** development phase ends Thursday 22 Oct, 00:59 WAT. Final phase: Thursday 22 Oct, 01:00–09:00 WAT. Only 3 uploads per day.

2. Where everyone stands right now

Team	Final score	Accuracy (F1)	Efik F1	Ibibio F1	Igala F1	Model size
datapsalm (1st)	0.539	0.586	0.14	0.03	0.16	87.6M
ray_nacido (2nd)	0.467	0.531	0.17	0.00	0.10	94.6M
us (godzilla)	first upload tonight	–	–	–	–	≈38M planned

Nobody is officially ranked yet: a team needs an F1 of at least 0.60 to be ranked. The two leaders are decent on the big languages and **almost blind on Efik, Ibibio and Igala**.

3. Six reasons we can win

1. **We know exactly where the points are.** Efik, Ibibio, Igala and Nupe carry **35% of the score**. The leader already averages about 0.77 on the other nine classes. Lifting just the four weak ones to 0.5 would take a team to about F1 0.68, well above the 0.60 ranking line and far ahead of today's 0.59.
2. **We know why everyone fails there.** In the training data each of these languages comes from only **1 to 3 people** (Nupe: one single speaker). A model learns those voices instead of the language, then fails on the new speakers in the test. A bigger model does not fix this: the second-placed team uses a strong African model and still scores 0.0 on Ibibio.
3. **We found a hidden trap in the data.** Each language mostly comes from one recording source with its own 'sound' (silence, loudness, microphone). A model can guess the label from that alone, scoring F1 0.34 **without hearing any speech**. Other teams' models will quietly learn this shortcut and break on phone audio. We erase it with targeted audio augmentation, so our model has to learn the language.

4. **We are adding new voices, legitimately.** Team-collected WhatsApp voice notes (phone audio, just like the test), an 8.8-hour Efik corpus, openly licensed Wikimedia Nigeria recordings, and, once the organisers approve, 10-speaker Igala, Nupe and Urhobo sets. Voice conversion then spreads many voices evenly across all languages.
5. **We are small and fast on purpose.** We start from AfriHuBERT, a model pretrained on 1,226 African languages including 11 of our 12, cut down to 4 layers (about 38M parameters). That earns an efficiency bonus of about ×0.94 against the leader's ×0.92, without giving up the language knowledge.
6. **We test honestly and package carefully.** Our own scorer reproduces the official score exactly. We validate on speakers the model has never heard, change one thing at a time, tune the final decisions to the exact scoring rule, and run the organisers' own checker before every upload. That way no submission is wasted and no surprise appears in the final phase.

4. What we are doing, step by step

We climb a ladder: start small, add one improvement per step, and keep it only if our speaker-held-out test says it helped. The biggest setup is only used if the team lead says so. Our public score climbs gradually; the strongest version is kept for the final phase.

Step	What changes	When (WAT)	Status
0	Smoke test: the whole pipeline works end to end	Tue midday	running
1	First real model: 4-layer AfriHuBERT, provided data only. **First upload (D1)**	Tue evening	next
2	Smarter decision rule for the small languages	Tue night	queued
3	Erase the recording 'fingerprints'; add phone, noise and echo	Tue night	queued
4	Extra legitimate data (Efik corpus, Wikimedia, Pidgin/English)	Wed morning	queued
5	Team voice notes + voice conversion + a 'teacher' model	Wed afternoon	queued
6	Try 6–8 layers if it pays off	Wed evening	queued
Final	Retrain the best recipe on all data; 3 final uploads	Thu 01:00–09:00	–

Training runs automatically in the cloud on team members' Kaggle GPUs, including overnight. Every result and every ready-to-upload submission is saved in our private GitHub repo.

5. How you can help today

- **Voice notes (most important).** Ask native speakers you know. Priority: **Nupe, Igala, Ibibio, Efik, Tiv** (10+ different people each), then Annang, Eggon, Idoma, Urhobo (5–10 each), plus Nigerian English and Pidgin. Each person records 3–5 minutes of natural talk, only in their language, and must be 18+.
- **Consent first.** Each person replies: "I agree that my voice recordings may be used to train and publish a language-identification model for the DSN 2026 competition. I am 18 or older." Also send first name, language, gender, age range and home town. More different people beats more minutes from one person.
- **Deadline:** Wednesday 21 October, 12:00 noon WAT. Send to the team's shared folder (the team lead will share the link).
- **Native-speaker checks:** we need one Urhobo speaker and one Efik/Ibibio/Annang speaker to listen to a few short clips.
- **Official members:** send your GitHub username for repo access, and keep your Kaggle account phone-verified.

6. Rules we never break

• Never use the three banned datasets (Hypa-Voices, Omnilingual ASR corpus, AfriSpeech IPA) or any copy of them; the test set came from them.
• Every outside dataset is listed in our model card. Grey-area sources wait for a written yes from the organisers.
• Teams are **at most 3 people** with one Codabench account. Helpers outside the three only recruit volunteers, pending the organisers' confirmation.
• One Kaggle account per person. No duplicate accounts.
• **Keep our methods and data sources inside the team.** Don't post them in public groups or forums.

7. Where to follow progress

- **STATUS.md** in our private GitHub repo (Godzilla-lab/dsn2026-lid): what is running, what is ready to upload, and the results of every step.
- **Full research report** (data sources, evidence, plan): research/ folder in the same repo.
- Questions: ask the team lead.